

ORIGINAL ARTICLE

Dental Flossing May Lower the Risk for Incident Ischemic Stroke, Cardioembolic Stroke Subtype, and Atrial Fibrillation

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BACKGROUND: Prior studies have demonstrated an association between oral infection and both incident stroke and atrial fibrillation (AF), a major cause of cardioembolic stroke. However, the impact of preventive oral behaviors such as dental flossing on incident stroke and AF remains unclear. We evaluated the relationship between flossing, incident stroke subtypes, and AF from the ARIC study (Atherosclerosis Risk in Communities).

METHODS: We conducted a prospective cohort study using data from the ARIC study. Flossing was assessed via a structured questionnaire at visit 4 (1996–1998). Participants were censored at the first stroke, death, or loss-to-follow-up through December 31, 2021. Incident ischemic stroke and subtypes were evaluated using Cox proportional hazards models, adjusting for cardiovascular risk factors, education, regular brushing, dental care utilization, periodontal disease, and dental caries.

RESULTS: Among 6200 dentate participants without prior stroke, 65% (n=4049) reported regular flossing. Flossers had significantly lower prevalence of vascular risk factors, periodontal disease, and dental caries at baseline. Flossers had lower IL (interleukin)-6 levels in crude analysis, but this difference was explained by confounding from age, body mass index, and smoking. CRP (C-reactive protein) levels did not differ between groups. During a median 23.7 years follow-up, 434 ischemic strokes occurred, including 146 thrombotic, 102 cardioembolic, 95 lacunar, and 91 other ischemic subtypes. Nonflossers had higher rates of both ischemic stroke (8.7% versus 6.1%) and nonstroke mortality (40.8% versus 34.5%). Flossing was associated with lower risk of ischemic stroke (adjusted hazard ratio, 0.77 [95% CI, 0.64–0.96]), cardioembolic stroke (adjusted hazard ratio, 0.60 [95% CI, 0.38–0.94]), and AF (adjusted hazard ratio, 0.88 [95% CI, 0.77–0.99]) with dose-response relationship (P trend=0.001) 6.6% of the effect mediated by lower AF incidence. Results remained robust in competing risk analysis (adjusted hazard ratio, 0.79 [95% CI, 0.63–0.98]) with an E value of 1.87.

CONCLUSIONS: Regular flossing was associated with a lower risk of ischemic stroke, primarily driven by reductions in cardioembolic stroke and AF, independent of other oral health behaviors.

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GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: atrial fibrillation ■ body mass index ■ cardiovascular disease ■ oral health ■ ischemic stroke

Oral diseases, such as periodontal disease (PD) and dental caries, are among the most prevalent non-communicable diseases globally.¹ Both are chronic

inflammatory conditions that affect not only oral health but are also associated with increased cardiovascular disease (CVD) and stroke risk.¹ PD is characterized by

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Nonstandard Abbreviations and Acronyms

AF	atrial fibrillation
ARIC	Atherosclerosis Risk in Communities
BMI	body mass index
CRP	C-reactive protein
CVD	cardiovascular disease
DARIC	dental ancillary study to ARIC
HR	hazard ratio
IL	interleukin
PA	physical activity
PD	periodontal disease

bacterial colonization, causing chronic inflammation and tissue damage.² Similarly, dental caries result from cariogenic bacteria producing acids that demineralize the tooth structure.³ Poor oral hygiene is a major contributor to both conditions, emphasizing the importance of oral hygiene practices as potential strategies to reduce CVD and stroke risk.

Dental flossing, alongside regular toothbrushing, is fundamental to good oral hygiene and is widely recommended for preventing PD and caries in both children and adults.⁴ Flossing targets interproximal plaque, a primary contributor to oral infection, and may reduce inflammation and bacterial loads that drive oral disease progression. Beyond maintaining oral health, growing evidence supports the role of oral self-care, such as brushing and flossing, in reducing systemic inflammation and lowering risk factors for CVD.⁵

A recent analysis in the Kuopio Oral Health and Heart study demonstrated that regular oral self-care practices can lead to significant survival benefits by reducing mortality related to atherosclerotic heart disease and stroke.⁶ These findings highlight the potential benefits of maintaining good oral hygiene, such as dental flossing.

Data from the US National Health and Nutrition Examination Survey demonstrate that daily flossing is associated with a reduced CVD events, stroke, and all-cause mortality,⁷ with higher frequencies of flossing showing greater risk reduction. Flossing has been linked to lower serum levels of high-sensitivity CRP (C-reactive protein), an established marker of systemic inflammation.⁷ Since inflammation contributes to atherosclerosis and CVD events, including myocardial infarction and stroke, reduced systemic inflammation through flossing provides a plausible mechanism for cardiovascular protection.

We recently reported the specific association between PD, ischemic stroke, and atrial fibrillation (AF), as well as a reduction in these outcomes associated with the regular dental care utilization.⁸ Similar associations were noted between caries and ischemic stroke, with a potential preventive effect of regular dental care utilization.⁹ Despite

these findings, gaps remain in our understanding of the specific effects of dental flossing on ischemic stroke, stroke subtypes, and AF. Additionally, the specific effects of flossing frequency on PD, caries, and systemic inflammatory markers, such as IL (interleukin)-6 and CRP, remain understudied. Hence, we aim to investigate the relationship between flossing and incident stroke, including its subtypes and AF, using data from the ARIC study (Atherosclerosis Risk in Communities). This large, longitudinal cohort provides a unique opportunity to examine these associations in a dentally well-characterized population. Additionally, we assessed the impact of flossing on markers of oral disease, such as PD and caries, as well as systemic inflammatory markers, including IL-6 and CRP. Our findings may have important implications for public health strategies aimed at reducing the burden of both oral and systemic diseases through accessible and cost-effective interventions like regular dental flossing.

MATERIALS AND METHODS

Data Availability Statement

The ARIC study data are available through the National Heart, Lung, and Blood Institute BioLINCC repository (<https://biolincc.nhlbi.nih.gov/studies/aric/>).

Study Population

The ARIC Study began in 1987 to examine the cause of atherosclerosis and its associated outcomes.¹⁰ Participants for this prospective cohort were enrolled from 4 separate communities in the US, including participants from Forsyth County, NC; Minneapolis, MN; Jackson, MS; and Washington County, MD. During visit 4 (1996–1998), regular tooth brushing, dental flossing, and dental care utilization were classified using patient-reported responses to the administered dental history questionnaire.¹¹ Dental flossing was assessed through the structured questionnaire that asked, how often did you use dental floss last week and the response was graded as "not at all", "1-time", "2 times", or "3 times or more". Those who reported not at all were classified as nonflossers, and all others as dental flossers. Subjects were reported to be regular brushing if they reported brushing their teeth $\geq 1 \times$ a day. They were classified as regular dental care users (those who sought routine dental care) or episodic users (those who sought dental care only when symptomatic). In eligible subjects (N=6483), dental examination was also conducted at visit 4, the dental ancillary study to ARIC (DARIC). The stroke and AF incident event data sets were evaluated for events through 2022 for 3 of the centers and up to 2020 for participants at the Jackson, MS center. and AF outcomes were assessed at follow-up through the end of 2022. The study design flow chart is depicted in Figure S1. Vascular risk factors for this analysis and education level were assessed during the visit. Education was divided into 3 categories: basic- less than completed high school, intermediate-completed high school or GED, and advanced- at least some college. The study team also contacted the ARIC study participants on an annual (pre-2012) and semiannual (post-2012) basis to evaluate for stroke.¹⁰ Participants provided written informed consent at each study visit. The 4 examination field

centers and the coordinating center were granted human subject approvals. The participants were followed for a median of 23.7 years. Hospital medical records were obtained, particularly if hospitalization occurred in this period. This study follows the STROBE reporting guideline.¹²

Assessment of Dental Caries and PD

Participants underwent full mouth caries assessments at visit 4 using modified Radike criteria.¹³ Presence of coronal or root caries was used as the primary exposure of interest. To further explore the dose-response relationship between dental caries and our outcomes, we calculated the decayed, missing, and filled surfaces index, which represents the total number of decayed, missing, and filled tooth/surfaces in the entire mouth, ranging from 0 (no caries) to 128. No distinction was made between teeth that were missing due to decay or missing due to another reason.¹⁴

Presence or absence of PD was also assessed as a covariate during visit 4 within DARIC. Details on the periodontal profile class may be found elsewhere.¹⁵ They were classified on a scale from periodontal profile class-A (periodontal health) or periodontal profile class-B through periodontal profile class-G (PD).¹⁵

Incident Ischemic Stroke, Stroke Subtype, and AF

Stroke events were identified using a computerized algorithm that screened hospital discharge records for *International Classification of Diseases, Ninth* codes 430-438, and *International Classification of Diseases, Tenth* codes I60-I69. Cases flagged by this algorithm or reported by participants during annual telephone follow-up underwent medical record abstraction by trained staff. A panel of physicians then adjudicated each potential stroke using standardized criteria from the National Survey of Stroke, which requires evidence of sudden or rapid onset of neurological symptoms lasting >24 hours or leading to death, with no apparent nonvascular cause. Transient ischemic attacks (*International Classification of Diseases, Ninth* code 435) were excluded from our outcome definition.

ARIC stroke surveillance is comprehensive, using: (1) hospital discharge codes from all local hospitals, (2) annual telephone follow-up to identify out-of-area events, and (3) surveillance of death certificates for stroke-related deaths. Stroke diagnoses were validated by physician review using standardized criteria, not based solely on administrative codes. Although our classification is modified from the TOAST criteria, it maintains the fundamental distinction between embolic and thrombotic mechanisms.¹⁶

Incident AF was assessed by 3 methods, including study EKGs, hospital discharge, *International Classification of Diseases, Ninth* CM codes 427.31 or 427.32, and death certificates.¹⁷ Standard, 10-second, 12-lead EKGs were obtained at baseline and at each of the subsequent follow-up examinations up until visit 5 (2011–2013). Tracings were performed in the supine position using MAC PC Personal Cardiographs (Marquette Electronics Inc) and transmitted electronically to the ARIC EKG Reading Center (Epidemiological Cardiology Research Center, Wake Forest School of Medicine, Winston Salem, NC), where they underwent automated reading and coding. Tracings with AF were reviewed by a cardiologist. Follow-up information was analyzed after visit 4 DARIC through calendar year 2019 for 3 of the sites and 2017 for the Jackson site, before the COVID-19 pandemic.

Inflammatory biomarkers (CRP and IL-6) were also collected from the ARIC study participants. IL-6 and high-sensitivity CRP concentrations were measured using ELISA and fit into a 7-point standard curve. Standard curve concentrations ranged from 0.156 to 10 pg/mL for serum IL-6 and 780 to 50 000 pg/mL for CRP.¹⁸

Based on the same DARIC data set, we reported an association between PD and incident AF, a possible mediator of ischemic stroke,⁸ which in turn increased risk of systemic ischemic events by almost 3×.¹⁹ Similarly, an association with periodontitis or dental plaque has been linked with AF.²⁰

Statistical Analysis

The final cohort consisted of participants without prior stroke who were assessed for dental flossing and underwent dental examination at visit 4. Demographics and covariates, including age, sex, race-center, body mass index (BMI), waist-to-hip ratio, hypertension, diabetes, education status, smoking status, alcohol use, cholesterol, and decayed, missing, and filled surfaces were compared using a *t* test for continuous variables and a χ^2 test for categorical variables. Hypertension was defined as systolic blood pressure ≥ 140 mmHg, diastolic blood pressure ≥ 90 mmHg, or current use of antihypertensive medications. Diabetes was defined as fasting glucose ≥ 126 mg/dL, nonfasting glucose ≥ 200 mg/dL, hemoglobin A1c $\geq 6.5\%$, or self-reported diabetes with medication use. We adjusted for race-center (a composite variable accounting for both race and ARIC study site, as recruitment strategies and population characteristics varied by center: White participants from Forsyth County, NC; White participants from Minneapolis suburbs, MN; White participants from Washington County, MD; Black participants from Jackson, MS; and Black participants from Forsyth County, NC).²¹ Triglyceride level, serum CRP, and IL-6 were not normally distributed and were assessed using a nonparametric Mann-Whitney *U* test. The Gray test was run to determine differences in the cumulative incidence for ischemic stroke between the dental flossers and nonflosser groups. The proportional hazards assumption was evaluated by including interaction terms between time and each covariate in the Cox models. Cox proportional hazards regression was used to estimate crude and adjusted hazard ratios (HR) and 95% CIs for incident ischemic stroke and its subtypes. These models were adjusted for age, gender, race-center, hypertension, diabetes, cholesterol, smoking, education, BMI, brushing frequency, and regular dental care use. Flossing frequency was modeled as an ordinal variable (0, 1, 2, and $\geq 3\times$ per week), with nonflossers as the reference group. A test for linear trend across flossing categories: none, irregular ($1\times$ – $2\times$ /wk), and regular ($\geq 3\times$ /wk) was conducted by including flossing frequency as an ordinal variable (0, 1, 2) in the Cox regression model and testing the significance of the coefficient. To examine whether flossing frequency was associated with baseline systemic inflammation, we performed regression analyses with log-transformed IL-6 and CRP as outcomes. Linear regression models examined the association between flossing frequency (coded ordinally: 0=none, 1=once/wk, 2=twice/wk, 3= $\geq 3\times$ /wk) and inflammatory markers in both unadjusted and adjusted models. Adjusted models included age, gender, BMI, education level, and smoking status as covariates. IL-6 and CRP were log-transformed to account for skewed distributions, with regression coefficients representing proportional changes. Subjects with missing data were excluded from analysis.

Competing Risk Analysis

We conducted Fine-Gray subdistribution hazard regression, treating death as a competing event. Sensitivity analyses included (1) restricting follow-up to the first 10 years after visit 4 to address behavior change; (2) additional adjustment for physical activity (PA) from visit 3 (1993–1995) and BMI from visit 4 (1996–1998); (3) loss-to-follow-up assessment. Loss-to-follow-up has been well-characterized in the ARIC cohort as predominantly attributable to mortality rather than voluntary withdrawal.²² The competing risk analysis was conducted using complete case analysis for all covariates, including PA, resulting in a slightly smaller analytical sample ($n=5516$) compared with the primary analysis ($n=6200$).

Sensitivity Analyses

Several sensitivity analyses confirmed the robustness of our findings. In an analysis restricted to individuals who reported any level of flossing (65%, excluding nonflossers), we compared regular flossers ($\geq 3\times/\text{wk}$ [$n=2916$], 46.8%) to irregular flossers ($1\times\text{--}2\times/\text{wk}$ [$n=1133$]; 18.2%), combining the once- and twice-weekly groups.

Mediation analyses were performed to determine the role of incident AF in mediating the effect of dental flossing on incident ischemic stroke, using structural equation modeling based on the Baron and Kenny framework.²³ We used the Valeri and VanderWeele²⁴ method to assess AF's role in mediating flossing's effect on stroke. Incidence rates were calculated from stroke events divided by person-time. Adjusted rates used multivariable Poisson regression with robust standard errors and log-person-time offset, with verified assumptions (deviance/ $\text{df}=0.94$).

From these models, we estimated marginal adjusted incidence rates for each exposure group (flossers and nonflossers) using predictive margins. Statistical significance of between-group differences in incidence rates was assessed using Wald tests. We calculated incidence rate ratios for ischemic stroke comparing flossers to nonflossers using multivariable Poisson regression models, with person-years of follow-up as the offset. Flossing status was assessed at baseline, and stroke events were ascertained during follow-up. Models were adjusted for age, gender, race-center, hypertension, diabetes, cholesterol, smoking, education, BMI, brushing frequency, and regular dental care use. Adjusted incidence rates were computed per 1000 person-years for each group.

To evaluate the robustness of the observed association to potential unmeasured confounding, we calculated E values for both the point estimate and the upper bound of the 95% CI. The E value quantifies the minimum strength of association that an unmeasured confounder would need to have with both flossing and ischemic stroke to explain away the observed association. All the data analyses for this study were conducted using SAS version 9.4 (Cary, NC).

RESULTS

The study enrollment and participant flow are shown in Figure 1. At visit 4, 11 365 individuals were assessed for floss status. Of those participants, 6453 dentate subjects were enrolled in DARIC (1317 refused oral examination, 1895 had medical contraindication to dental exam at the time (dentist or physician ever told you that you needed to take antibiotics before every dental visit, congenital heart disease, rheumatic heart disease, heart murmur, endocarditis, mitral valve prolapse, pacemaker, heart, kidney or other organ transplant, artificial heart valve, stent, shunt or artificial joint, kidney dialysis, major surgery, radiation or chemotherapy within the last 2 months, taking prednisone or an immunosuppressive medication) and 1619 were edentulous, comprising the major reasons for exclusion.²¹ Of these, 6200 study participants without prior ischemic stroke had complete covariate information and underwent full mouth periodontal assessment of 6 sites per tooth as well as a dental caries exam. Compared with excluded participants, included participants were younger (62.3 versus 63.4 years), had lower cardiovascular risk factors (hypertension, 33.0% versus 44.9%; diabetes, 13.2% versus 21.1%; current smoking, 12.0% versus 18.2%), and higher educational attainment (44.3% versus 32.2% with advanced education; all $P<0.001$), indicating selection toward a healthier population subset (Table S1).

At visit 4, individuals included in the DARIC study were divided into 2 groups: dental flossers ($n=4049$) compared with nonflossers ($n=2151$; Table). Flossers were more likely to be female, White, have lower BMI, cardiovascular risk factors (hypertension, diabetes, smoking),

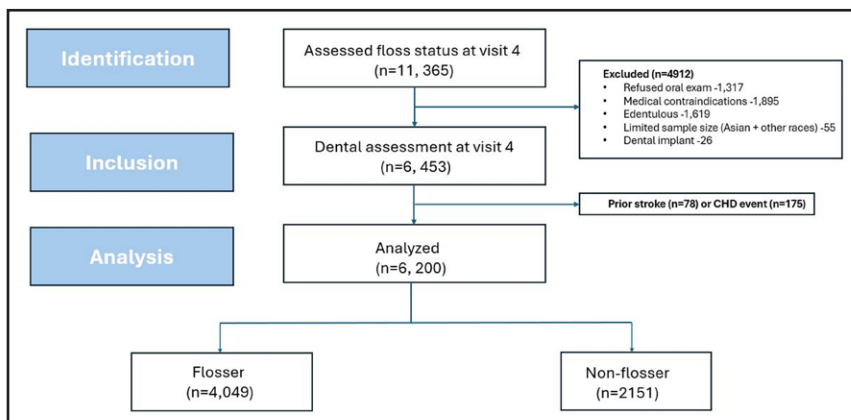


Figure 1. STROBE diagram for incident ischemic stroke.

Table. Characteristics of the Study Participants, According to Dental Flossing Status

	Dental flossing		
	Flossers (n=4049)	Nonflossers (n=2151)	P value
Age, y*	62.3±5.6	62.4±5.6	0.374
Sex (%)			
Female	62.3	40.0	<0.001
Male	37.7	60.0	
Race (%)			
Black	13.9	26.1	<0.001
White	86.1	73.9	
BMI*	28.2±5.2	29.4±5.5	<0.001
Waist-to-hip ratio*	0.94±0.07	0.96±0.07	<0.001
Hypertension (%)	31.5	36.0	<0.001
Diabetes (%)	11.7	16.0	<0.001
Education (%)			
Basic	8.6	20.1	<0.001
Intermediate	43.6	42.2	
Advanced	47.8	37.7	
Smoking (%)			
Never	50.0	43.0	<0.001
Former	39.5	42.2	
Current	10.5	14.9	
Alcohol (%)			
Never	18.1	20.2	<0.001
Former	24.5	29.6	
Current	57.4	50.2	
Cholesterol, mg/dL*	202.3±35.8	198.7±35.9	<0.001
LDL*	121.7±32.9	122.3±33.1	0.489
HDL*	52.2±16.9	47.9±16.2	<0.001
Triglycerides, mg/dL†	123 (88–171)	122 (86–173)	0.578
High sensitivity CRP, mg/dL‡	2.7 (1.0–7.0)	2.7 (0.9–6.8)	0.321
IL-6, pg/mL‡	1.4 (0.7–2.3)	1.5 (0.8–2.7)	<0.001
DMFS	67±25	70±28	<0.001
Caries present	13.4	35.2	<0.001
Periodontal disease	65.2	85.1	<0.001
Ischemic stroke	6.1	8.7	<0.001
Thrombotic	1.9	2.9	0.002
Cardioembolic	1.2	2.2	
Lacunar	1.5	1.5	
Others	1.5	2.1	
Hemorrhagic stroke	0.3	0.2	0.446
Incident AF	20.1	22.0	0.084

DMFS index was calculated as the total number of decayed (D), missing (M), and filled (F) tooth/surfaces. Periodontal disease (PPC-B through G) and periodontal health (PPC-A). AF indicates atrial fibrillation; BMI, body mass index; CRP, C-reactive protein; DMFS, decayed, missing, and filled surfaces; HDL, high-density lipoprotein; IL, interleukin; LDL, low-density lipoprotein; and PPC, periodontal profile class.

*Mean±SD.

†Median (25–75 percentile).

‡CRP levels available in 3377 flossers and 1780 nonflossers, IL-6 levels available in 3241 flossers, and 1720 nonflossers.

alcohol use, and higher education. They also had slightly higher total cholesterol and a higher level of HDL cholesterol. They had better oral health (lower PD and caries) and slightly lower IL-6 levels, though CRP did not differ significantly. This healthier baseline profile may contribute to reduced stroke risk. The number of participants in the flosser group who suffered an ischemic stroke was n=246, and the number of participants in the nonflosser group who suffered an ischemic stroke was n=188, a total of 434 stroke events. Of the 434 ischemic strokes, there were 146 thrombotic, 102 cardioembolic, 95 lacunar, and 91 other subtypes.

Additionally, we further tested the association between flossing frequency and systemic inflammatory markers. In unadjusted analysis, flossing frequency showed a modest inverse association with log-transformed IL-6 levels ($B=-0.017$ [95% CI, -0.025 to -0.008]; $P<0.001$), with each category increase in flossing frequency associated with a 1.7% decrease in IL-6. However, after adjusting for age, gender, BMI, education level, and smoking status, this association was completely attenuated and no longer statistically significant ($B=-0.007$ [95% CI, -0.015 to 0.002]; $P=0.134$). In the fully adjusted model for IL-6, the strongest independent predictors were BMI ($B=0.014$; $P<0.001$), age ($B=0.005$; $P<0.001$), and smoking status ($B=0.029$; $P=0.014$). The adjusted model explained 4.3% of the variance in log IL-6 levels ($R^2=0.043$, $F[6,4525]=33.817$; $P<0.001$). No association was observed between flossing frequency and CRP levels in either unadjusted ($B=0.008$ [95% CI, -0.004 to 0.019]; $P=0.190$) or adjusted analyses ($B=0.002$ [95% CI, -0.009 to 0.014]; $P=0.672$). In the fully adjusted CRP model, the strongest predictors were BMI ($B=0.026$; $P<0.001$) and gender ($B=-0.242$; $P<0.001$). The adjusted model explained 12.2% of the variance in log CRP levels ($R^2=0.122$, $F[6,4467]=103.130$; $P<0.001$).

The cumulative incidence curve of the 2 risk groups can be seen in Figure 2. Significant divergence in incidence curves was observed between groups stratified by flossing status, emerging around 3 years post-visit 4, as indicated by the Gray test ($P<0.001$). However, dental flossing still met the proportional hazards assumption using the time interaction criteria. During a median of 23.7 years (interquartile range, 19.8–25.2) follow-up period, the flosser group had a significantly lower risk of ischemic stroke (HR, 0.62 [95% CI, 0.51–0.75]) compared with the nonflosser group. The significantly lower risk (adjusted HR, 0.77 [95% CI, 0.64–0.96]) remained after adjustment for age, gender, race-center, hypertension, diabetes, hypercholesterolemia, smoking status, 3-level education, regular brushing, and regular dental care user (Figure 3). In looking at the individual subtypes, the above association was largely driven by the flosser group having a significantly lower risk of cardioembolic stroke (HR, 0.50 [95% CI, 0.33–0.75]) that remained significant

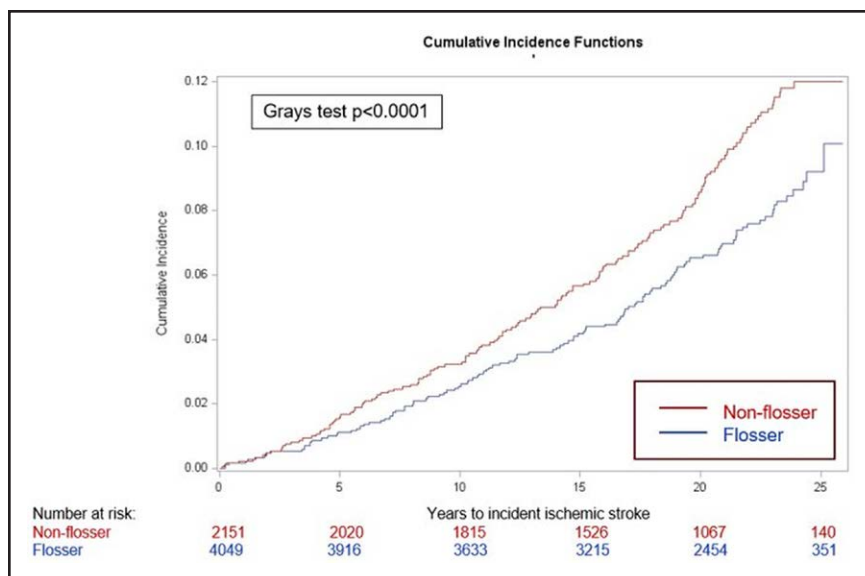


Figure 2. Cumulative incidence curve.

after adjustments for confounders listed above (adjusted HR, 0.60 [95% CI, 0.38–0.94]; Table S2). Although the flosser group had a significantly lower risk of thrombotic stroke (HR, 0.61 [95% CI, 0.44–0.86]), this was not significant after adjustments for confounders listed above (adjusted HR, 0.89 [95% CI, 0.61–1.31]). Flossing was not associated with lowering of lacunar or other ischemic stroke subtypes in the crude or adjusted model. No significant association was observed between flossing and hemorrhagic stroke (adjusted HR, 0.45 [95% CI, 0.15–1.31]). When analyzing the composite outcome of any stroke (ischemic or hemorrhagic, N=530 events), regular flossing remained protective (adjusted HR, 0.78 [95% CI, 0.65–0.95]).

There was also a noted threshold between frequency of flossing and reduced ischemic stroke reduction based on whether 1 reported flossing ≥ 3 /wk, 2/wk, 1/wk compared with nonflossers (Figure 4). The test for linear trend was statistically significant (P for trend=0.001), although the association was primarily driven by the contrast between nonflossers and any flossing.

Although the trend was statistically significant, the association was primarily driven by the difference between nonflossers and any flossing. Among individuals who reported flossing, there was no significant association between flossing frequency and stroke risk. Compared with those who flossed once a week (reference

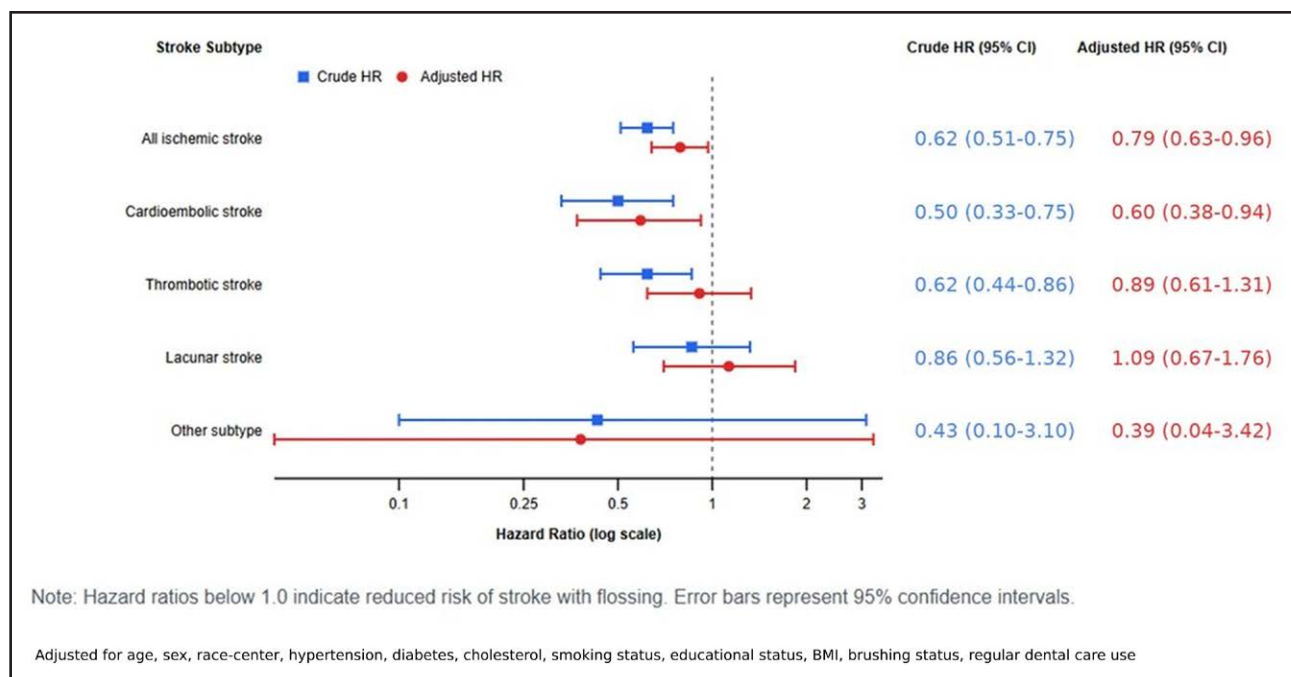


Figure 3. Crude and adjusted* hazard ratio (HR) for ischemic stroke and stroke subtypes. BMI indicates body mass index.

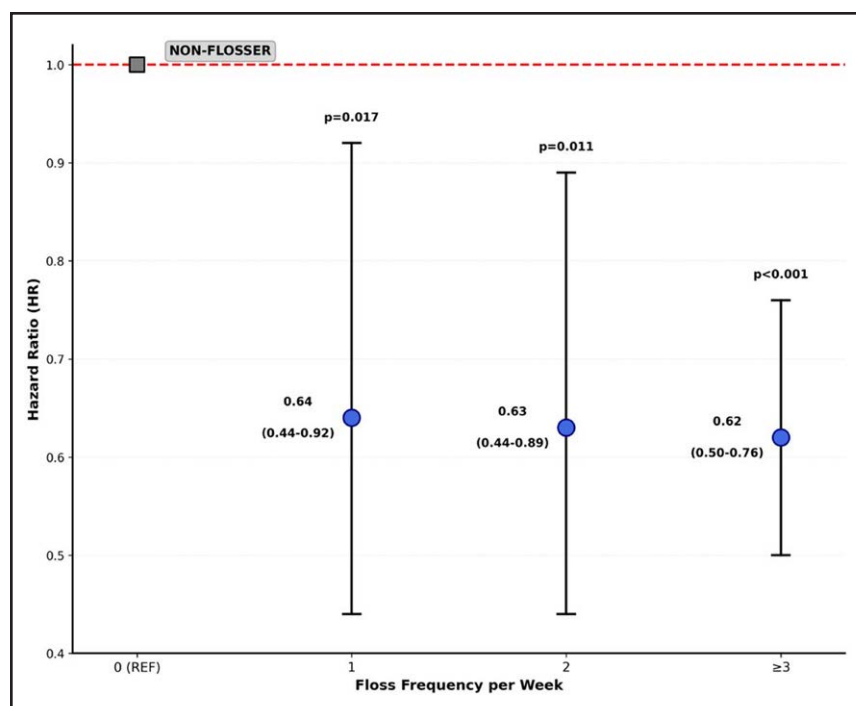


Figure 4. Impact of flossing frequency on stroke risk.

group), individuals who flossed twice a week had an HR of 0.98 (95% CI, 0.61–1.58; $P=0.944$), while those who flossed 3 or more times per week had a HR of 0.95 (95% CI, 0.66–1.38; $P=0.791$). The overall comparison across flossing frequencies among flossers was also not significant ($P=0.956$).

For mediation analysis, only AF events occurring before stroke were considered, ensuring appropriate temporal sequence. In Cox models adjusting for AF, we treated AF as a time-varying covariate to account for changes in AF status during follow-up. Mediation analysis revealed AF accounted for only 6.6% of the flossing-stroke association, suggesting other pathways predominate. Given the modest mediation by AF, potential mechanisms include (1) reduction in systemic inflammation; (2) decreased endothelial dysfunction; (3) reduced atherosclerotic burden; (4) prevention of bacterial translocation and subsequent vascular injury. Unfortunately, biomarkers to formally test these pathways were unavailable at Visit 4.

As shown in Figure 5, regular flossing was associated with a lower risk of incident AF (HR=0.83), and reduced incident AF was, in turn, associated with a reduced risk of ischemic stroke (HR=0.42). The total effect of flossing on ischemic stroke yielded an HR of 0.61. When adjusting for incident AF, the direct effect of flossing on stroke was attenuated (HR=0.63), suggesting partial mediation.

Crude incidence rates of ischemic stroke were significantly lower among flossers (5.25 per 1000 person-years [95% CI, 4.31–6.39]) compared with nonflossers (6.82 per 1000 person-years [95% CI, 5.98–7.78]), with a statistically significant difference ($P<0.0001$).

After adjustment for age, sex, race-center, hypertension, diabetes, cholesterol, smoking, alcohol use, education, and obesity, the incidence rate among flossers remained similar at 5.25 per 1000 person-years (95% CI, 4.36–6.32), while the rate among nonflossers increased to 6.82 per 1000 person-years (95% CI, 5.69–8.18). The adjusted difference between groups remained statistically significant ($P=0.010$), supporting a persistent association between flossing and reduced stroke risk even after accounting for major cardiovascular and demographic risk factors. In adjusted analyses, individuals who reported regular flossing had a 23% lower incidence rate of ischemic stroke compared with nonflossers (incident rate ratio=0.77 [95% CI, 0.64–0.92]). The adjusted incidence rate was 5.25 per 1000 person-years among flossers and 6.82 per 1000 person-years among nonflossers (Figure S2). In our competing risk analysis with additional adjustment for PA (do you exercise or play sports?) assessed on visit 3 (1993–1995), and BMI from visit 4 (1996–1998) showed a persistent association (HR, 0.79 [95% CI, 0.63–0.98]). PA showed an independent protective effect (adjusted HR, 0.79 [95% CI, 0.64–0.99]), demonstrating that flossing and PA have parallel, independent associations with stroke risk. Combined with the moderate E value for the lower confidence limit (1.37), these findings suggest that while unmeasured confounding cannot be entirely excluded, the observed association is unlikely to be fully explained by residual confounding (Figure S3). We acknowledge that residual confounding from unmeasured factors such as income, diet quality, and other health-seeking behaviors may exist.

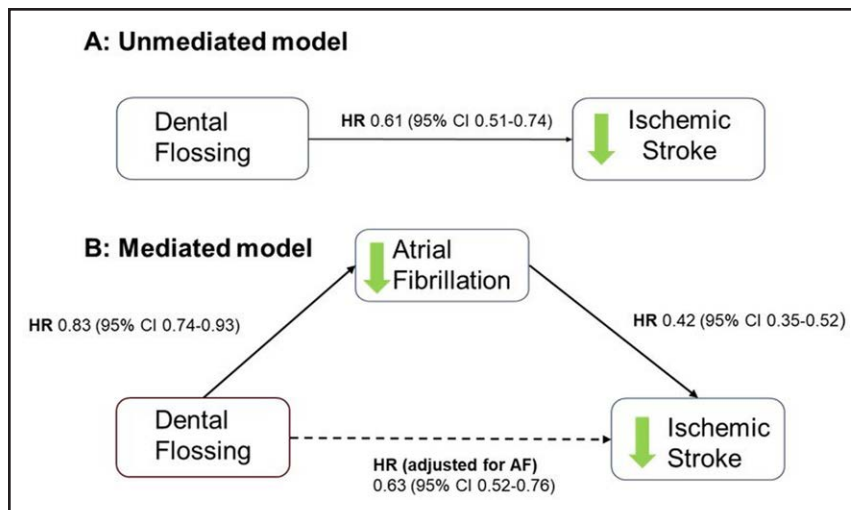


Figure 5. Mediation analysis testing the role of atrial fibrillation (AF) in mediating the effect of flossing on stroke risk reduction. HR indicates hazard ratio.

The *E* value for the lower confidence limit (1.37) indicates moderate robustness to unmeasured confounding, though factors such as diet quality or health consciousness could potentially attenuate the association. The *E* value for the upper CI bound (1.31) suggests modest confounders could shift results toward the null, indicating the association is relatively robust but not immune to unmeasured confounding.

DISCUSSION

In this prospective cohort study of 6200 ARIC participants followed for a median of 23.7 years, regular dental flossing was associated with a 23% lower risk of ischemic stroke (adjusted HR, 0.77 [95% CI, 0.64–0.96]), particularly cardioembolic stroke (adjusted HR, 0.60 [95% CI, 0.38–0.94]). AF partially mediates this association (6.6%). A statistically significant trend was observed ($P=0.001$); however, the association reflected a threshold effect, with lower risk among those reporting any flossing compared with none, rather than a graded frequency-dependent response. Associations with lacunar or thrombotic stroke were not significant after adjustment.

Dental flossers ($n=4049$) were compared with non-flossers ($n=2151$). Flossers had a healthier baseline profile, including lower cardiovascular risk factors, less alcohol use, higher education levels, and better oral health (lower PD and caries incidence). Although flossers had slightly lower IL-6 levels in crude analysis, multivariable regression demonstrated this difference was explained by confounding from age, BMI, education, and smoking status, suggesting flossing serves as a marker of overall health-conscious behavior rather than independently affecting systemic inflammation. CRP levels did not differ between groups.

The competing risk analysis strengthens our findings by demonstrating that the protective association of flossing with ischemic stroke persists when accounting

for the competing risk of death (subdistribution HR, 0.77 [95% CI, 0.62–0.97]). Importantly, flossing was not associated with nonstroke mortality (subdistribution HR, 0.96 [95% CI, 0.86–1.06]), suggesting the observed benefit is specific to cerebrovascular outcomes rather than reflecting a general health selection effect. PA showed an independent protective effect (adjusted HR, 0.79 [95% CI, 0.64–0.99]), demonstrating that flossing and PA have parallel, independent associations with stroke risk. This specificity supports a mechanistic pathway linking oral hygiene practices to stroke prevention, potentially through reduction in systemic inflammation and endothelial dysfunction, or other vascular pathways, rather than simply serving as a marker of overall health-conscious behavior.

Building on these observations, the lower IL-6 levels among flossers, in conjunction with better oral health indicators, may reflect reduced systemic inflammatory burden linked to improved vascular health and lower AF risk. AF is increasingly recognized as an inflammation-mediated condition in which immune cell infiltration and inflammatory mediators alter atrial electrophysiology, disrupt calcium homeostasis, impair connexin function, and promote heterogeneous conduction.²⁵ These processes, coupled with myolysis, cardiomyocyte apoptosis, and activation of fibrotic pathways via TGF- β and MMP, contribute to atrial structural remodeling and heightened AF susceptibility. Inflammation also plays a central role in thromboembolism development, a major AF complication. Although CRP levels did not differ significantly between flossers and nonflossers, the IL-6 divergence may indicate that flossing selectively suppresses pro-inflammatory pathways most relevant to atrial remodeling and thromboembolic risk. This selective attenuation could partially mediate the observed reduction in cardioembolic stroke, reinforcing the hypothesis that targeted oral hygiene practices may influence cardiovascular outcomes through modulation of AF-associated inflammatory pathways.

Our findings are reinforced by recent national data from the National Health and Nutrition Examination Survey (2009–2016), in which daily flossing was associated with lower prevalence of CVD events and reduced CVD mortality (HR, 0.64 [95% CI, 0.49–0.84]), independent of metabolic risk factors and sociodemographic confounders.⁷ Studies, including a 2024 US population-based study, have shown that flossing less than twice weekly is linked to higher all-cause mortality, while daily flossing is associated with better CRP profiles, reflecting reduced low-grade inflammation.²⁵

Additionally, data from National Health and Nutrition Examination Survey 2017 to 2020 support our results: regular flossing was associated with a 24% lower likelihood of atherosclerotic CVD (adjusted odds ratio, 0.76 [95% CI, 0.60–0.97]), and a 32% lower risk of stroke or myocardial infarction using stringent definitions (odds ratio, 0.68 [95% CI, 0.49–0.94]).²⁶ A clear dose-response was again observed, with flossing ≥ 3 days per week associated with greater risk reduction. Stratified analysis confirmed the benefit was consistent across sex, diabetes, and hyperlipidemia subgroups.

Results from the Kuopio Oral Health and Heart study further strengthen these associations. In a prospective cohort with 18.8 years of follow-up, brushing and flossing were associated with significantly reduced CVD mortality (HR, 0.25 [95% CI, 0.07–0.89]), while brushing alone was not (HR, 0.72 [95% CI, 0.37–1.41]).⁶ This underscores the added value of flossing beyond brushing alone. Notably, mouthwash use provided no additional benefit in reducing CVD mortality, aligning with our focus on mechanical plaque disruption as the primary beneficial mechanism.

Limitations include assessment of flossing at a single time point without objective validation, potentially introducing nondifferential misclassification that would bias results toward the null. However, our 10-year sensitivity analysis demonstrated continued significant association (adjusted HR, 0.84 [95% CI, 0.72–0.99]), and flossing is a relatively stable learned behavior in adults. The healthier profile of included versus excluded participants may underestimate the true association. Single dental exam, use of education as socioeconomic proxy, and exclusion of non-Black/non-White participants limit generalizability and measurement granularity.

The specificity of the association to flossing, but not other oral hygiene behaviors, deserves consideration. Although this pattern could reflect residual confounding from unmeasured health behaviors, several factors support a more direct relationship. First, flossing uniquely targets interproximal plaque removal in areas inaccessible to brushing, which harbor periodontal pathogens implicated in systemic inflammation.²⁷ Second, the association's specificity to cardioembolic stroke (and AF) but not lacunar stroke supports a mechanism beyond general cardiovascular health. Third, the threshold effect (any

flossing versus none, rather than a frequency gradient) is inconsistent with flossing being a simple marker of health consciousness, which would more likely show a clear dose-response. Nevertheless, we cannot exclude the possibility that flossing serves as a marker for other unmeasured protective factors.

Despite extensive covariate adjustment and sensitivity analyses, residual confounding from unmeasured factors (diet quality, health consciousness, social determinants) remains possible, though the *E* value (1.37) suggests moderate robustness. The consistency of this relationship across the National Health and Nutrition Examination Survey and Kuopio Oral Health and Heart studies further supports its robustness. Future studies should investigate whether specific inflammatory or arrhythmogenic pathways directly mediate this link.

This may be one of the first US-based studies to prospectively demonstrate a significant independent association between dental flossing and reduced incident ischemic stroke. Our findings support the importance of flossing as a modifiable oral health behavior linked to cardiovascular and cerebrovascular risk. Integrating flossing recommendations into cardiovascular prevention frameworks, particularly for AF and cardioembolic stroke, may yield significant public health benefits.

ARTICLE INFORMATION

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Supplemental Material

Tables S1–S2

Figures S1–S3

REFERENCES

- Jain N, Dutt U, Radenkovic I, Jain S. WHO's Global Oral Health Status Report 2022: actions, discussion and implementation. *Oral Dis*. 2024;30:73–79. doi: 10.1111/odi.14516
- Pihlstrom BL, Michalowicz BS, Johnson NW. Periodontal diseases. *Lancet*. 2005;366:1809–1820. doi: 10.1016/S0140-6736(05)67728-8
- Qin X, Zi H, Zeng X. Changes in the global burden of untreated dental caries from 1990 to 2019: a systematic analysis for the Global Burden of Disease study. *Heliyon*. 2022;8:e10714. doi: 10.1016/j.heliyon.2022.e10714
- Worthington HV, MacDonald L, Poklepovic Pericic T, Sambunjak D, Johnson TM, Imai P, Clarkson JE. Home use of interdental cleaning devices, in addition to toothbrushing, for preventing and controlling periodontal diseases and dental caries. *Cochrane Database Syst Rev*. 2019;4:CD012018. doi: 10.1002/14651858.CD012018.pub2
- Frisbee SJ, Chambers CB, Frisbee JC, Goodwill AG, Crout RJ. Association between dental hygiene, cardiovascular disease risk factors and systemic inflammation in rural adults. *J Dent Hyg*. 2010;84:177–184
- Janket SJ, Lee C, Surakka M, Jangam TG, Van Dyke TE, Baird AE, Meurman JH. Oral hygiene, mouthwash usage and cardiovascular mortality during 18.8 years of follow-up. *Br Dent J*. 2023;1–6. doi: 10.1038/s41415-023-5507-4
- Philip N, Tamimi F, Al-Sheebani A, Almuzafar A, Shi Z. The effect of self-reported flossing behavior on cardiovascular disease events and mortality: findings from the 2009–2016 National Health and Nutrition Examination Surveys. *J Am Dent Assoc*. 2025;156:17–27.e3. doi: 10.1016/j.adaj.2024.09.017
- Sen S, Redd K, Trivedi T, Moss K, Alonso A, Soliman EZ, Magnani JW, Chen LY, Gottesman RF, Rosamond W, et al. Periodontal Disease, Atrial Fibrillation and Stroke. *Am Heart J*. 2021;235:36–43. doi: 10.1016/j.ahj.2021.01.009
- Sen S, Logue L, Logue M, Otersen EAL, Mason E, Moss K, Curtis J, Hicklin D, Nichols C, Rosamond WD, et al. Dental caries, race and incident ischemic stroke, coronary heart disease, and death. *Stroke*. 2024;55:40–49. doi: 10.1161/STROKEAHA.123.042528
- The Atherosclerosis Risk in Communities (ARIC) Study: design and objectives. The ARIC investigators. *Am J Epidemiol*. 1989;129:687–702.
- ARIC. Dental procedures manual: baseline examination 1996–1999 (version 1.0, June 7, 1996). 1996
- von EE, Altman DG, Egger M, Pocock SJ, Gotsche PC, Vandenbroucke JP, Initiative S. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. *J Clin Epidemiol*. 2008;61:344–349. doi: 10.1016/j.jclinepi.2007.11.008
- Radike A. *Criteria for the diagnosis of dental caries*. American Dental Association; 1968:88
- Wang X, Shaffer JR, Zeng Z, Begum F, Vieira AR, Noel J, Anjomshoa I, Cuenco KT, Lee MK, Beck J, et al. Genome-wide association scan of dental caries in the permanent dentition. *BMC Oral Health*. 2012;12:57. doi: 10.1186/1472-6831-12-57
- Morelli T, Moss KL, Beck J, Preisser JS, Wu D, Divaris K, Offenbacher S. Derivation and validation of the periodontal and tooth profile classification system for patient stratification. *J Periodontol*. 2017;88:153–165. doi: 10.1902/jop.2016.160379
- Rosamond WD, Folsom AR, Chambless LE, Wang CH, McGovern PG, Howard G, Copper LS, Shahar E. Stroke incidence and survival among middle-aged adults: 9-year follow-up of the Atherosclerosis Risk in Communities (ARIC) cohort. *Stroke*. 1999;30:736–743. doi: 10.1161/01.str.30.4.736
- Alonso A, Agarwal SK, Soliman EZ, Ambrose M, Chamberlain AM, Prineas RJ, Folsom AR. Incidence of atrial fibrillation in whites and African-Americans: the Atherosclerosis Risk in Communities (ARIC) study. *Am Heart J*. 2009;158:111–117. doi: 10.1016/j.ahj.2009.05.010
- Beck JD, Moss KL, Morelli T, Offenbacher S. Periodontal profile class is associated with prevalent diabetes, coronary heart disease, stroke, and systemic markers of C-reactive protein and interleukin-6. *J Periodontol*. 2018;89:157–165. doi: 10.1002/JPER.17-0426
- Shi M, Chen LY, Bekwelem W, Norby FL, Soliman EZ, Alam AB, Alonso A. Association of atrial fibrillation with incidence of extracranial systemic embolic events: the ARIC study. *J Am Heart Assoc*. 2020;9:e016724. doi: 10.1161/JAHA.120.016724
- Struppek J, Schnabel RB, Walther C, Heydecke G, Seedorf U, Lamprecht R, Smeets R, Borof K, Zeller T, Beikler T, et al. Periodontitis, dental plaque, and atrial fibrillation in the Hamburg City Health Study. *PLoS One*. 2021;16:e0259652. doi: 10.1371/journal.pone.0259652
- Elter JR, Champagne CM, Offenbacher S, Beck JD. Relationship of periodontal disease and tooth loss to prevalence of coronary heart disease. *J Periodontol*. 2004;75:782–790. doi: 10.1902/jop.2004.75.6.782
- George KM, Folsom AR, Kucharska-Newton A, Mosley TH, Heiss G. Factors related to differences in retention among African American and White Participants in the Atherosclerosis Risk in Communities Study (ARIC) prospective cohort: 1987–2013. *Ethn Dis*. 2017;27:31–38. doi: 10.18865/ed.27.1.31
- Baron RM, Kenny DA. The moderator-mediator variable distinction in social psychological research: conceptual, strategic, and statistical considerations. *J Pers Soc Psychol*. 1986;51:1173–1182. doi: 10.1037//0022-3514.51.6.1173
- Valeri L, Vanderweele TJ. Mediation analysis allowing for exposure-mediator interactions and causal interpretation: theoretical assumptions and implementation with SAS and SPSS macros. *Psychol Methods*. 2013;18:137–150. doi: 10.1037/a0031034
- Yu YH, Steffensen B, Chasman DI, Buring JE. Self-reported oral health is associated with systemic health outcomes and all-cause mortality. *J Am Dent Assoc*. 2024;155:233–243.e8. doi: 10.1016/j.adaj.2023.11.006
- Fei J, Gong X. Association between dental floss use and atherosclerotic cardiovascular disease in American adults. *Heart Lung*. 2025;71:32–38. doi: 10.1016/j.hrtlng.2025.02.003
- Hu YF, Chen YJ, Lin YJ, Chen SA. Inflammation and the pathogenesis of atrial fibrillation. *Nat Rev Cardiol*. 2015;12:230–243. doi: 10.1038/nrcardio.2015.2